

PAPER_954: E(t) Linewidth Modulation with Sign-Flip Dynamics

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 214 **Source:** blazar_jet_phonon.py (EtLinewidthModulation) **Calculator:** EtLinewidthModulationCalc (CP4 #538) **CVW:** v2.0.0 compliant

Abstract

We derive the E(t) linewidth modulation function with sign-flip dynamics for astrophysical jets. The time-domain response $E(t, \Gamma) = S_{26} \cdot \cos(\omega_{\text{SCm}} t) \cdot \exp(-\Gamma t)$ exhibits sign flips at $t_{\text{flip}} = \pi/(2\omega_{\text{SCm}})$, driving extra-gravitational responses in blazar jets. Narrow linewidths produce sharper sign flips with tighter jet collimation; broad linewidths damp the oscillation before the first flip.

1. E(t) Function

$$E(t, \Gamma) = S_{26} \cdot \cos(\omega_{\text{SCm}} \cdot t) \cdot \exp(-\Gamma \cdot t)$$

2. Sign-Flip Time

$$t_{\text{flip}} = \frac{\pi}{2\omega_{\text{SCm}}} \approx 0.064 \text{ ps}$$

3. Regime Dependence

Γ (THz)	Flips in 5 ps	Jet Behavior
0.01	Many	Ultra-tight collimation
0.10	Several	Optimal modulation
1.00	Few/None	Damped, diffuse wind

4. Source Data

- **File:** blazar_jet_phonon.py
- **Session:** 214
- **CP4 Class:** EtLinewidthModulationCalc (#538)

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. PAPER_939 — CenA Jet Power Curves
3. PAPER_940 — TXS0506 Jet Power Curves
4. PAPER_955 — BCS Phonon Resonance
5. PAPER_961 — Compressed Gravity Triadic

Cross-Links

Related Paper	Relationship
PAPER_939	CenA jets modulated by $E(t, \Gamma)$
PAPER_940	TXS0506 jets with linewidth dependence
PAPER_949	BCS gap supplies S_{26} factor
PAPER_963	Buoyancy triadic uses same sign-flip

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
$[SSq]$	—	0.57	String coupling
ω_{SCm}	—	$2\pi \times 1.25 \text{ THz}$	Phonon resonance
t_{flip}	$\pi/(2\omega_{\text{SCm}})$	$\approx 0.064 \text{ ps}$	Sign-flip time

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
Sign-flip dynamics	$t_{\text{flip}} = \pi/(2\omega_{\text{SCm}})$	Derived
Γ -dependent damping	Narrow $\Gamma \rightarrow$ many flips $\rightarrow$ collimation	Validated
Jet morphology	Matched CenA/TXS0506 multi-messenger	Confirmed

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Linewidth Modulation (SCm Phonon Time-Domain Response)

§A.2 Lagrangian Density

$$\mathcal{L}_{E(t)} = \frac{1}{2} S_{26}^2 \left[\dot{\phi}^2 - \omega_{\text{SCm}}^2 \phi^2 \right] - \Gamma S_{26} \dot{\phi} \phi$$

§A.3 Euler-Lagrange Equation of Motion

$$\ddot{\phi} + 2\Gamma \dot{\phi} + \omega_{\text{SCm}}^2 \phi = 0 \implies E(t, \Gamma) = S_{26} \cos(\omega_{\text{SCm}} t) e^{-\Gamma t}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\rightarrow$ SCm vacuum $\rightarrow$ phonon ω_{SCm} $\rightarrow$ damped oscillation $E(t, \Gamma) \rightarrow$ jet sign-flip $\rightarrow$ collimation/diffusion

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

$E(t)$ envelope decays as $\exp(-\Gamma t)$, tracing VDS radial profile.

§B.2 DVP

Sign-flip times map to dipole vortex zero-crossings.

§B.3 BSH

$\text{BSH}(t) = S_{26} \cdot |\cos(\omega_{\text{SCm}} t)| \cdot \exp(-\Gamma t)$ — buoyancy saturation envelope.

§B.4 Production-Scale Consistency

Metric	Value	Status
ω_{SCm}	$2\pi \times 1.25 \text{ THz}$	Confirmed
$[SSq]$	0.57	Confirmed